

Assignment-4

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Subject: Statistical Method and Probability theory

Section: I

Q.1

Define:

(i) Statistical Hypothesis:-

A hypothesis is a conjecture about the parameters of population

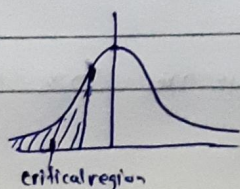
(ii) Null Hypothesis and Alternate Hypothesis

* Null Hypothesis (H_0):- A hypothesis which is actually tested for acceptance or rejection is known as null hypothesis. It contains a statement of equality ('=', '<=', '≥').

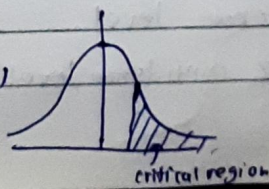
* Alternate Hypothesis (H_1 or H_a):- A statement of population parameter which gives an alternate to null hypothesis i.e. if H_0 is to be accepted, H_1 has to be rejected and vice versa. It contains a statement of inequality ('≠', '<', '>').

(iii) One tailed and two tailed test

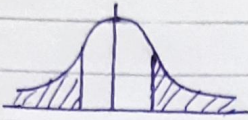
* One tailed test:- A type of hypothesis test in which the alternate hypothesis (H_1) is testing for a possibility of an effect in only one direction (contains either greater than or less than sign)



(or)



- * two tailed test:- A hypothesis test in which the alternate hypothesis (H_1) tests for the possibility of an effect in both directions (Contains not equal to sign ($\neq$))



(iv) Type I and Type II error.

- * Type I error (α):- The error of rejecting the Null Hypothesis (H_0), when H_0 is true.
- * Type II error (β):- The error of accepting the Null Hypothesis (H_0), when alternate hypothesis (H_1) is true.

(v) Significance level and Critical region.

- * Significance level (α):- It is the quantity of risk of type I error which we are ready to tolerate in making a decision about null hypothesis (H_0)
- * Critical Region:- The values of test-statistics on which the null hypothesis is rejected is known as the "Critical Region" or "Rejection Region".

(vi) Confidence interval and confidence limits.

- * Confidence interval (CI):- A confidence interval is a range of values used to estimate the true value of a population parameter. It is Based on sample data and is associated with confidence level.
- Eg. for 95% confidence level, we are 95% confident that

the true parameter lie within the interval.

* Confidence limits: are two values that define the bounds of a confidence interval.

Q.2. Statistical hypothesis under test is called Null Hypothesis.

Q.3. A wrong decision about the null hypothesis leads to Type I and Type II types of errors.

Q.4. The probability of type I error is given by α (level of significance).

Q.5. The level of significance of a test is related to type I error and is given by α .

Q.6. The critical region provides us the criterion for rejection of null hypothesis.

Q.7. Which one of the following statement is true for Type II error

Ans. (b) Accepting an incorrect Hypothesis

Q.8. An investigator commits a type II error when he/she
Ans. (d) accepts a null hypothesis when it is false.

Q. 9. Type I error in the testing of null hypothesis occurs when the researchers

Ans. (a) rejects it, when
~~(b)~~

Q-10 (A) The heights of 10 adult males selected at random from a given locality had a mean of 158 cm and variance 39.0625 cm. State the null and alternate Hypothesis.

Sol. let population mean = μ_0 ($\because \mu_0$ is not given)
~~do~~ Null hypothesis, $H_0: \mu = \mu_0$
 Alternate hypothesis, $H_1: \mu \neq \mu_0$

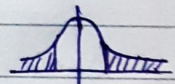
(B) A local telephone company claims that the average length of a phone call is 8 minutes. In a random sample of 18 phone calls, the sample mean was 7.8, $S = 0.5$ minutes. Is there enough evidence to support claim at $\alpha = 0.05$?

Sol. Since hypothesis for mean, i.e. we can use t-test

* Stating Null & alternate hypothesis:

$H_0: \mu = 8$ (given claim)

$H_1: \mu \neq 8$ (i.e. two tailed test)



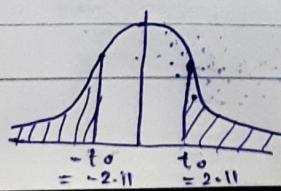
* $\alpha = 0.05$

* degree of freedom $= n - 1 = 18 - 1 = 17$

(given sample size $(n) = 18$)

* Critical values for $\alpha = 0.01$ (two tailed), for degree of freedom = 17

$t_0 = \pm 2.110$



* test statistics :-

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

given, sample mean $\bar{x} = 7.8$

sample size $n = 18$

sample standard deviation $s = 0.5$

population mean $\mu = 8$

$$\therefore t = \frac{7.8 - 8}{\frac{0.5}{\sqrt{18}}} = -1.7$$

Since t lies in non-critical region, we do not have enough reason to reject the claim.

Q-11(A) A travel agency's marketing brochure indicates that the standard deviations of hotel room rates for two cities are the same. A random sample of 13 room rates in one city has a standard deviation of \$27.50 and a random sample of 16 hotel room rates in the other city has a standard deviation of \$29.75. State the null and alternate hypothesis.

Sol $\Rightarrow$ If σ_1^2 and σ_2^2 are variance of hotel room rates in city 1 and City 2 respectively.

* The Null Hypothesis :-

$$H_0: \sigma_1^2 = \sigma_2^2$$

* The Alternate hypothesis:

$$H_1: \sigma_1^2 \neq \sigma_2^2$$

(B) Consider the following measurements of the heat produced by two mines:

Mine 1	8260	8130	8350	8070	8340	
Mine 2	7950	7890	7900	8140	7840	7920

State the Null and Alternate hypothesis.

Sol

$$\mu_1 = \frac{8260 + 8130 + 8350 + 8070 + 8340}{5} = 8230$$

$$\mu_2 = \frac{7950 + 7890 + 7900 + 8140 + 7840 + 7920}{6} = 7940$$

Sol

Null hypothesis, $H_0: \mu_1 > \mu_2$

Alternate hypothesis, $H_1: \mu_1 \leq \mu_2$

Q-12 (A)

A manufacturer claims that its rechargeable batteries have an average life greater than 1000 charges. A random sample of 10 Batteries has a mean life of 1002 charges and standard deviation of 14.

State the NULL and alternate hypothesis

Sol

Null hypothesis $\Rightarrow H_0: \mu = 1000$

Alternate hypothesis $\Rightarrow H_1: \mu > 1000$

(B)

Certain pesticide is packed into bags by a machine. A random sample of 10 bags is drawn and their contents are found to weight in kgs as follows: 50, 49, 44, 52, 45, 48, 46, 48, 49, 45. State the null and alternate hypothesis that the average packing can be taken as 50kg.

Sol. Null hypothesis (H_0) : $\mu = 50 \text{ kg}$
Alternate hypothesis (H_1) : $\mu \neq 50 \text{ kg}$

Q-13 Given the frequency function, $f(x, \theta) = \frac{1}{\theta}$, $0 \leq x \leq \theta$ and $f(x, \theta) = 0$, otherwise and that you are testing the null hypothesis $H_0: \theta = 1$, against $H_1: \theta = 2$, by means of a single observed value x . What would be the sizes of the Type I and type II errors, if you choose the interval

(i) $0.5 \leq x$

(ii) $1 \leq x \leq 1.5$ as the critical regions?

Sol (i) if critical Region: $0.5 \leq x$

* Type I error (α): -

$$\alpha = P\{x \in W | H_0\} \quad (\text{where } W \text{ is critical region})$$

$$\alpha = P\{0.5 \leq x | \theta = 1\}$$

$$\alpha = P(0.5 \leq x \leq \theta)$$

$$\alpha = P(0.5 \leq x \leq 1)$$

$$\alpha = \int_{0.5}^1 \frac{1}{\theta} d\theta \Big|_{\theta=1}$$

$$\alpha = \int_{0.5}^1 \frac{1}{1} d\theta$$

$$\alpha = 0.5 \quad \text{Ans}$$

* Type II error size (β): -

$$\beta = P\{x \in \bar{W} | H_1\}$$

$$\beta = P\{x < 0.5 | \theta = 2\}$$

$$\beta = P(0 \leq x < 0.5)$$

{ since given $0 \leq x \leq \theta$, ie }
lower limit of $x = 0$

$$\beta = \int_0^{0.5} \frac{1}{\theta} d\theta \Big|_{\theta=2}$$

$$\beta = \int_0^{0.5} \frac{1}{2} d\theta$$

$$\beta = 0.25 \quad \text{Ans}$$

(ii) critical region: $1 \leq x \leq 1.5$

* Type I error size (α):-

$$\alpha = P(X \in W | H_0)$$

$$\alpha = P(1 \leq x \leq 0.5 | \theta = 1)$$

$$\alpha = P(1 \leq x \leq 1) \quad \text{as } (0 \leq x \leq \theta)$$

$$\alpha = \int_1^1 \frac{1}{\theta} d\theta \Big|_{\theta=1}$$

$$\alpha = \int_1^1 \frac{1}{1} d\theta$$

$$\alpha = 0$$

* Type II error size (β):-

$$\beta = P(X \in W | H_1)$$

$$\beta = 1 - P(X \in W | H_1)$$

$$\beta = 1 - P(1 \leq x \leq 1.5 | \theta = 2)$$

$$\beta = 1 - P(1 \leq x \leq 1.5)$$

$$\beta = 1 - \int_1^{1.5} \frac{1}{\theta} d\theta \Big|_{\theta=2}$$

$$\beta = 1 - \int_1^{1.5} \frac{1}{2} d\theta$$

$$\beta = 1 - \frac{1}{2} [1.5 - 1]$$

$$\beta = 1 - \frac{0.5}{2}$$

$$\beta = 0.75$$

Q14. Let p be the probability that a coin will fall head in a single toss in order to test $H_0: p = \frac{1}{2}$ against $H_1: p = \frac{3}{4}$. The coin is tossed 5 times and H_0 is rejected if more than 3 heads are obtained. Find the probability

of type I error & type II error

Sol =

Null Hypothesis: $H_0: p = \frac{1}{2}$

Alternate Hypothesis: $H_1: p = \frac{3}{4}$

$$P(X=k) = {}^n C_k p^k (1-p)^{n-k}$$

given $n=5$,

type I error

$$\alpha = P(X > 3 | p = \frac{1}{2}) = P(X=4) + P(X=5)$$

$$\alpha = {}^5 C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^1 + {}^5 C_5 \left(\frac{1}{2}\right)^5$$

$$\alpha = 5 \times \frac{1}{32} + \frac{1}{32}$$

$$\alpha = \frac{3}{16} \quad \underline{\text{Ans}}$$

type II error

$$\beta = P(X \leq 3 | p = \frac{3}{4}) = P(X=0) + P(X=1) + P(X=2) + P(X=3)$$

$$\beta = {}^5 C_0 \left(\frac{3}{4}\right)^0 \left(1-\frac{3}{4}\right)^5 + {}^5 C_1 \left(\frac{3}{4}\right)^1 \left(1-\frac{3}{4}\right)^4$$

$$+ {}^5 C_2 \left(\frac{3}{4}\right)^2 \left(1-\frac{3}{4}\right)^3 + {}^5 C_3 \left(\frac{3}{4}\right)^3 \left(1-\frac{3}{4}\right)^2$$

$$\beta = 1 \times 1 \times \frac{1}{1024} + \frac{5 \times 3}{4} \times \frac{1}{256}$$

$$+ \frac{10 \times 9}{16} \times \frac{1}{64} + \frac{10 \times 27}{64} \times \frac{1}{16}$$

$$\beta = \frac{47}{128} \quad \underline{\text{Ans}}$$

Q-15

If $x \geq 1$ is the critical region for testing $H_0: \theta = 2$ against the alternative $\theta = 1$, on the basis of the single observation from the population.

$$f(x, \theta) = \theta \exp(-\theta x), \quad 0 \leq x < \infty$$

Obtain the values of type I and type II errors

Soln

Critical region $W: x \geq 1$

$\bar{W}: x < 1$

* Type I error (α):

$$\begin{aligned} \alpha &= P(x \in W | H_0) \\ &= P(x \geq 1 | \theta = 2) \\ &= P(1 \leq x < \infty) \\ &= \int_1^{\infty} \theta e^{-\theta x} dx \Big|_{\theta=2} \\ &= \int_1^{\infty} 2 e^{-2x} dx \\ &= \left[\frac{2e^{-2x}}{-2} \right]_1^{\infty} \\ &= [-e^{-2x}]_1^{\infty} \\ &= -e^{-\infty} + e^{-2} \\ &= 0 + e^{-2} \end{aligned}$$

$$\alpha = \frac{1}{e^2} \quad \text{Ans}$$

* Type II error (β):

$$\begin{aligned} \beta &= P(x \in \bar{W} | H_1) \\ &= P(x < 1 | \theta = 1) \\ &= P(0 \leq x < 1) \end{aligned}$$

$$\begin{aligned}
 &= \int_0^1 \theta e^{-\theta x} dx \Big|_1 \\
 &= \int_0^1 1 e^{-1x} dx \\
 &= \left[\frac{e^{-x}}{-1} \right]_0^1 \\
 &= -e^{-1} + e^0 \\
 &= 1 - \frac{1}{e} \\
 &\beta = (e-1)/e
 \end{aligned}$$

Q-16. A random variable X follows the following exponential distribution

$$f(x, \theta) = \frac{1}{\theta} e^{-x/\theta}, \quad 0 \leq x < \infty, \quad \theta > 0$$

The null hypothesis $H_0: \theta = 2$ is rejected. ^(alternate hypothesis $H_1: \theta = 4$) If the observation selected at random takes the value 6 or more, find the critical region, acceptance region, and the type I and type II error.

Sol.

$$f(x, \theta) = \frac{1}{\theta} e^{-x/\theta}, \quad 0 \leq x < \infty, \quad \theta > 0$$

Null hypothesis: $H_0: \theta = 2$

Alternate hypothesis: $H_1: \theta = 4$

Critical Region: $X \geq 6$

Acceptance Region: $X < 6$

Type I error (α):

$$\alpha = P\{X \in W | H_0\}$$

$$\alpha = P\{X > 6 | \theta = 2\}$$

$$\alpha = P(6 \leq X < \infty)$$

$$\alpha = \int_6^{\infty} \frac{1}{2} e^{-x/2}$$

$$\alpha = \left[\frac{1}{2} \frac{e^{-x/2}}{(-\frac{1}{2})} \right]_6^{\infty}$$

$$\alpha = \left[-e^{-x/2} \right]_6^{\infty}$$

$$\alpha = -e^{-\infty} + e^{-6/2}$$

$$\alpha = e^{-3}$$

$$\alpha = 0.0497$$

Type II error (β):

$$\beta = P(X \in \bar{W} | H_1)$$

$$\beta = P(X < 6 | \theta = 4)$$

$$\beta = P(0 \leq X < 6)$$

$$\beta = \int_0^6 \frac{1}{4} e^{-x/4}$$

$$\beta = \left[\frac{1}{4} \frac{e^{-x/4}}{(-1/4)} \right]_0^6$$

$$\beta = \left[-e^{-x/4} \right]_0^6$$

$$\beta = -e^{-6/4} + e^0$$

$$\beta = 1 - e^{-1.5}$$

$$\beta = 0.7768$$